

REV 2026

Semester I

Lab Course

1.5 Credits

45 Hours

# Engineering Drawing with CAD

Course Code 1004 — Complete Polytechnic Revision 2026 Handbook

PROGRAMME(S)

**AI, AI & ML, Biomedical Engg, Cloud & Big Data, Cyber Forensics, CS & Tech, CS & Engg, Computer Engg, CS & Engg (AI & ML), Electronics & Comm, Electronics Engg, Electronics Engg (Embedded Systems), Electronics & Computer Engg, IC Design & Fabrication, Instrumentation Engg, IT, Micro Electronics, Automation & Robotics, RPA, Textile Tech**

SUBJECT CODE

**1004**

SEMESTER

**I**

COURSE TYPE

**Lab**

TEACHING SCHEME (L:T:P:R)

**0:0:3:0**

CONTACT HOURS

**45**

CREDITS

**1.5**

CIE MARKS

**60**

ESE MARKS

**40**

# Official Source Declaration

 **Source:** Kerala Polytechnic Revision 2026 Syllabus — Course 1004 (Engineering Drawing with CAD). This handbook is prepared strictly from the official syllabus document provided. All module headings, topics, subtopics, practical outcomes, assessment schemes, and learning resources are derived directly from the source.

 **Verified Inconsistencies:** The syllabus lists the Teaching Scheme as 0:0:3:0 (L:T:P:R) with 45 contact hours and 1.5 credits. The assessment table shows CIE 60 and ESE 40. The self-learning description in the detailed syllabus table (Page 2) is partially cut off in the PDF. The open-ended experiment description states that drawing sheets from Modules I and II may be considered open-ended experiments. The series test schedule is noted as "Series test - I (module II/II)" and "Series test- II (module II/II)" — the module numbering appears to use Roman numerals inconsistently (II vs II). This handbook treats "module II" as Module II and "module III/IV" as Modules III/IV respectively, following the context. The CO-PO matrix is presented with correlation levels; this handbook reproduces it as given. All other content is faithfully extracted.

## Course Dashboard

45

CONTACT HOURS

1.5

CREDITS

4

MODULES

60+40

CIE + ESE

### Course Introduction

Engineering graphics is the fundamental language of engineers used to visually communicate ideas, designs, and technical information without any language barriers. Its objectives include developing the ability to interpret and create accurate drawings. Understanding the shape, size, and relationship of components, and applying standards for clear communication. It also enhances spatial visualization and supports the effective planning and manufacture of engineering products. Introduction of CAD software and practicing the development of

drawings along with Engineering graphics give an insight to students regarding development of drawings with accuracy and efficiency without paper usage in engineering design.

## Pre-requisites

- Basic Geometry (Secondary Education)

### Teaching & Assessment Scheme

| L | T | P | S | Self-learning (SS)/Week | Total Hrs/Sem | Credits C | CIE | ESE | Total |
|---|---|---|---|-------------------------|---------------|-----------|-----|-----|-------|
| 0 | 0 | 3 | 0 | 1.5                     | 45            | 1.5       | 60  | 40  | 100   |

L: Lecture (1 credit), T: Tutorial (1 credit), P: Practical (0.5 credit), S: Project (0.5 credit), SS: Self-Learning, CIE: Continuous Internal Evaluation, ESE: End Semester Examination

## How to Use This Handbook

### 1

#### Understand

Read each module's concepts, definitions, and principles.

### 2

#### Visualise

Study the diagrams, SVGs, and worked examples to see the geometry in action.

### 3

#### Apply

Practice with the module-wise questions, calculators, and CAD exercises.

### 4

#### Revise

Use the quick-revision cards, formula bank, and model paper before exams.

 **Tip:** Use the Print / Save as PDF button to save a printable PDF of the entire handbook. The Search box at the bottom helps you quickly find topics across all modules.

## Course Outcomes & CO-PO Mapping

### Course Objectives

1. To familiarize the language of graphics which is used to express ideas, convey instructions while carrying out engineering jobs.
2. Provide skills to translate ideas into sketches and to draw and read various engineering curves, conic sections and the generation of 3D solid surfaces into plane areas.
3. To apply basic skill to develop projection of points, lines, orthographic views and sectional views of objects.
4. To familiarize the use of CAD software.
5. To acquire skill in developing orthographic and sectional views using CAD.

# Course Outcomes (CO)

## Course Outcomes and Bloom's Taxonomy Level

| CO  | Course Outcome                                                                           | Bloom's Level |
|-----|------------------------------------------------------------------------------------------|---------------|
| CO1 | Construct geometrical figures and illustrate development of surfaces.                    | Apply         |
| CO2 | Interpret projections of points and lines, orthographic projections and sectional views. | Apply         |
| CO3 | Familiarization in using CAD software.                                                   | Apply         |
| CO4 | Develop orthographic projections and sectional views in CAD.                             | Apply         |

# CO-PO Mapping

## Course Articulation Matrix

| Course Outcomes | PROGRAM OUTCOMES |     |     |     |     |     |     |     |     |      |      |
|-----------------|------------------|-----|-----|-----|-----|-----|-----|-----|-----|------|------|
|                 | PO1              | PO2 | PO3 | PO4 | PO5 | PO6 | PO7 | PO8 | PO9 | PO10 | PO11 |
| CO1             | 3                | 2   | 3   | -   | -   | -   | -   | -   | -   | -    | -    |
| CO2             | 3                | 3   | 2   | -   | -   | -   | -   | -   | -   | -    | -    |
| CO3             | 3                | -   | 2   | 3   | -   | -   | -   | -   | -   | -    | -    |
| CO4             | 3                | 2   | 3   | 3   | -   | -   | -   | -   | -   | -    | -    |

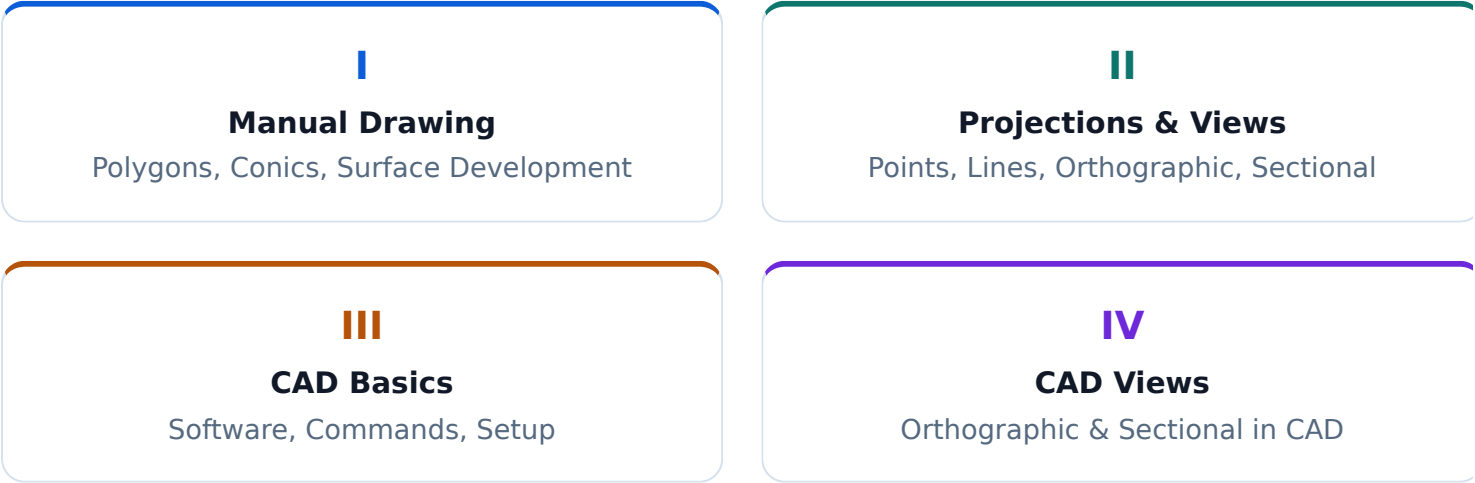
Legends: 1 - Low; 2 - Medium; 3 - High; No Correlation - “-”

# Curriculum Coverage Map

## Module-wise Topics, Hours, CO & Handbook Anchor

| Module | Topics                                                                           | Hours | CO  | Anchor     |
|--------|----------------------------------------------------------------------------------|-------|-----|------------|
| I      | Regular Polygons, Conic Sections, Development of Surfaces                        | 9     | CO1 | Module I   |
| II     | Basic Projections, Projections of Points & Lines, Orthographic & Sectional Views | 15    | CO2 | Module II  |
| III    | CAD Software Familiarization & Basic Commands                                    | 9     | CO3 | Module III |
| IV     | Orthographic & Sectional Views in CAD                                            | 9     | CO4 | Module IV  |

# How the Modules Connect



MODULE I

# Drawing — Regular Polygons, Conic Sections & Development of Surfaces

Learn to construct regular polygons, draw conic sections (ellipse, parabola, hyperbola), and develop surfaces of 3D solids into 2D patterns.

Hours: 9

CO: CO1

Bloom: Apply

Type: Manual Drawing

## 1.1 Regular Polygons — Construction of Pentagon & Hexagon▼

**Definition:** A regular polygon is a closed plane figure with all sides equal in length and all interior angles equal. Common regular polygons used in engineering drawing include pentagon (5 sides), hexagon (6 sides), heptagon (7 sides), and octagon (8 sides).

**Key Terminology:**

- **Side (s):** Length of one side.
- **Interior angle:**  $(n - 2) \times 180^\circ / n$  where n is the number of sides.
- **Exterior angle:**  $360^\circ / n$
- **Circumradius (R):** Radius of the circle passing through all vertices.
- **Apothem (r):** Distance from centre to midpoint of a side.

For a regular polygon with n sides and side length s:  
Circumradius  $R = s / (2 \times \sin(180^\circ/n))$  | Apothem  $r = s / (2 \times \tan(180^\circ/n))$

### Construction of a Regular Pentagon (Side Length Given)

**Method: General method using circumcircle**

1. Calculate the circumradius using  $R = s / (2 \times \sin(36^\circ))$ .
2. Draw a circle with radius R and mark the centre O.
3. Draw a vertical diameter AB.
4. Divide the circle into 5 equal parts ( $72^\circ$  each) using a protractor or compass.
5. Join the consecutive points to form the pentagon.

**WORKED EXAMPLE**

**Given:** Side length = 40 mm. Construct a regular pentagon.

$R = 40 / (2 \times \sin 36^\circ) = 40 / (2 \times 0.5878) = 40 / 1.1756 \approx 34.03 \text{ mm.}$

Draw circle of radius 34.03 mm, divide into 5 equal parts ( $72^\circ$ ), join the points.

## Construction of a Regular Hexagon (Side Length Given)

### Method: Using circumcircle

1. Calculate  $R = s$  (since  $\sin 30^\circ = 0.5$ ,  $R = s / (2 \times 0.5) = s$ ).
2. Draw a circle with radius equal to the given side length.
3. Mark any point on the circle as the first vertex.
4. Using the same radius, step off 6 equal divisions around the circle ( $60^\circ$  each).
5. Join the points to get the hexagon.

#### WORKED EXAMPLE

**Given:** Side length = 30 mm. Construct a regular hexagon.

Since  $R = s = 30$  mm, draw a circle of radius 30 mm. Step off 6 divisions of 30 mm along the circumference. Join the points.



Regular Pentagon (left) and Hexagon (right) constructed using the circumcircle method. The dashed circle shows the circumcircle.

 **Tip:** For a hexagon, the side length equals the circumradius — this is a useful shortcut. Always label all vertices and show construction arcs lightly in pencil.

**Malayalam support:** സമാന്തരകോണികൾ (Regular Polygons) — സമാന്തരകോണികൾ സമാന്തരകോണികൾ സമാന്തരകോണികൾ. സമാന്തരകോണികൾ (5 സമാന്തരകോണികൾ), സമാന്തരകോണികൾ (6 സമാന്തരകോണികൾ) സമാന്തരകോണികൾ സമാന്തരകോണികൾ. സമാന്തരകോണികൾ (circumradius) സമാന്തരകോണികൾ സമാന്തരകോണികൾ, സമാന്തരകോണികൾ  $R = s/(2 \sin 36^\circ)$ , സമാന്തരകോണികൾ  $R = s$  സമാന്തരകോണികൾ.

**Definition:** Conic sections are curves obtained by intersecting a right circular cone with a plane. The three main conic sections are:

- **Ellipse:** Plane cuts all generators of the cone (angle between plane and axis  $>$  semi-vertical angle).
- **Parabola:** Plane is parallel to one generator (angle = semi-vertical angle).
- **Hyperbola:** Plane cuts both nappes (angle  $<$  semi-vertical angle).

### Ellipse — Rectangular Method

This method is used when the major and minor axes are given.

1. Draw the major axis AB and minor axis CD intersecting at O.
2. Construct a rectangle with sides equal to AB and CD.
3. Divide the rectangle into equal parts horizontally and vertically.
4. Draw lines from the ends of the axes through the division points.
5. Mark the intersections and draw a smooth curve through them.

### Ellipse — Concentric Circle Method

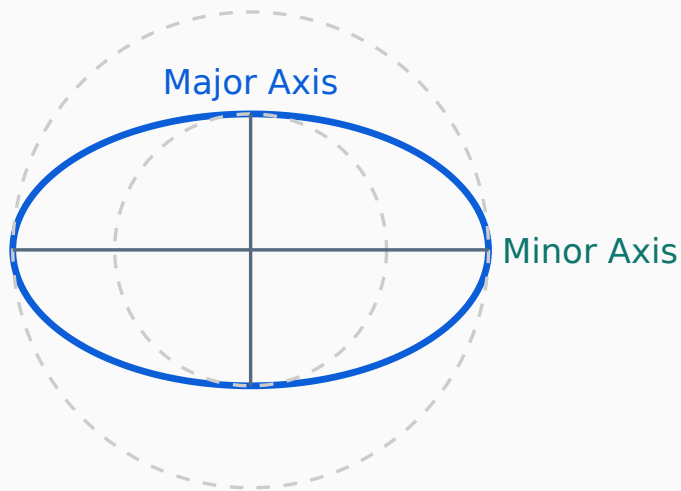
This method uses two concentric circles with diameters equal to the major and minor axes.

1. Draw two concentric circles with diameters equal to the major axis (outer) and minor axis (inner).
2. Divide the circles into 12 equal parts ( $30^\circ$  intervals).
3. From each division on the outer circle, draw a vertical line downward.
4. From each division on the inner circle, draw a horizontal line outward.
5. The intersections of these lines give points on the ellipse.
6. Join the points with a smooth curve.

#### WORKED EXAMPLE — CONCENTRIC CIRCLE METHOD

**Given:** Major axis = 80 mm, Minor axis = 50 mm.

Draw circles of radii 40 mm and 25 mm. Divide into 12 parts. Project vertically from outer circle and horizontally from inner circle. Intersection points trace the ellipse.



### Concentric Circle Method — Ellipse

Ellipse construction using the concentric circle method. The outer circle radius = half major axis, inner circle radius = half minor axis.

**Tip:** The ellipse is a closed curve with two axes of symmetry. The sum of distances from any point on the ellipse to the two foci is constant and equal to the major axis length.

**Malayalam support:** കോൺ സെക്ഷൻസ് (Conic Sections) — കോൺ സെക്ഷൻസ് കോൺ സെക്ഷൻസ്. കോൺ സെക്ഷൻസ് (Ellipse), പാരബോള (Parabola), ഹൈപ്പർബോള (Hyperbola) കോൺ സെക്ഷൻസ്. കോൺ സെക്ഷൻസ് കോൺ സെക്ഷൻസ് കോൺ സെക്ഷൻസ്.

## 1.3 Development of Surfaces — Rectangular Prism & Cylinder▼

**Definition:** Development of a surface is the process of unfolding a 3D solid's surface into a 2D plane (a net). It is used in sheet metal work, packaging, and fabrication to determine the exact shape and size of material needed.

**Key Principle:** The developed surface must have the same area as the original surface, and the edges of the development correspond to the edges of the solid.

### Development of a Rectangular Prism

A rectangular prism (cuboid) has 6 rectangular faces. Its development consists of 4 side rectangles arranged in a strip, plus 2 end rectangles (top and bottom).

1. Draw the four side faces in a row: each face has height = prism height, width = corresponding side length.
2. Attach the top and bottom faces to the appropriate edges.



3. Add fold lines (dashed) and tabs if required.

### WORKED EXAMPLE

**Given:** Prism dimensions: length = 50 mm, width = 30 mm, height = 40 mm.

Development: Four rectangles of size 50×40, 30×40, 50×40, 30×40 arranged in a strip.  
Top and bottom: 50×30 each attached to the top and bottom edges.

## Development of a Cylinder

A cylinder consists of a curved lateral surface and two circular bases.

- **Lateral surface:** Develops into a rectangle of height = cylinder height and width = circumference =  $2\pi r$ .
- **Bases:** Two circles of radius  $r$  attached to the top and bottom edges of the rectangle.

$$\text{Lateral surface area} = 2\pi rh \quad | \quad \text{Total surface area} = 2\pi r(r + h)$$

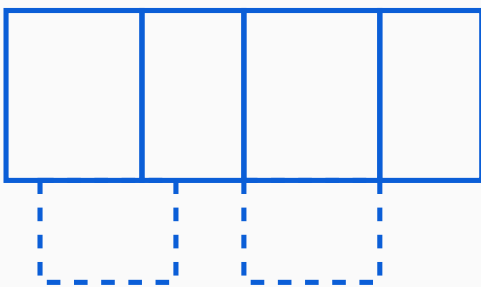
### WORKED EXAMPLE

**Given:** Cylinder radius  $r = 20$  mm, height  $h = 60$  mm.

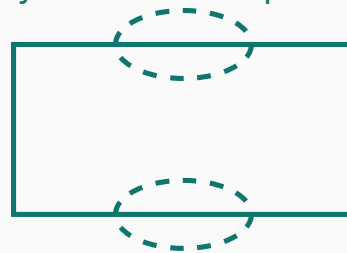
Lateral development: rectangle of width (circumference) =  $2\pi \times 20 = 125.66$  mm, height = 60 mm.

Bases: two circles of radius 20 mm.

Prism Development



Cylinder Development



$$\text{Rectangle} = 2\pi r \times h \quad | \quad \text{Circles} = \pi r^2$$

Development (net) of a rectangular prism (left) showing side faces and top/bottom faces, and a cylinder (right) showing lateral rectangle and circular bases.



**Malayalam support:** പരമാവർത്തം വികാസം (Development of Surfaces) — 3D പരമാവർത്തം വികാസം 2D പരമാവർത്തം വികാസം. പരമാവർത്തം വികാസം പരമാവർത്തം പരമാവർത്തം വികാസം. പരമാവർത്തം വികാസം (prism) പരമാവർത്തം 6 പരമാവർത്തം, പരമാവർത്തം വികാസം പരമാവർത്തം 2 പരമാവർത്തം വികാസം.



## Drawing – Basic Projections, Points, Lines, Orthographic & Sectional Views

Understand first-angle and third-angle projections, draw projections of points and lines, and develop orthographic and sectional views of engineering objects.

**Hours:** 15

**CO:** CO<sub>2</sub>

**Bloom:** Apply

**Type:** Manual Drawing

## 2.1 Theory of Projection — First Angle & Third Angle▼

**Definition:** Projection is the method of representing a 3D object on a 2D plane by projecting its points along parallel lines (orthographic projection) or from a point (perspective projection). Engineering drawing uses **orthographic projection**.

## Two main systems:

- **First Angle Projection (European system):** Object is placed in the first quadrant (between the observer and the planes). The view is projected onto the plane behind the object.
- **Third Angle Projection (American system):** Object is placed in the third quadrant (between the planes and the observer). The view is projected onto the plane in front of the object.

### Comparison of First Angle and Third Angle Projection

| Feature         | First Angle                      | Third Angle                      |
|-----------------|----------------------------------|----------------------------------|
| Object position | Between observer and plane       | Between plane and observer       |
| View placement  | Front view above, top view below | Front view below, top view above |

| Feature       | First Angle                    | Third Angle               |
|---------------|--------------------------------|---------------------------|
| Symbol        | Concentric circles (European)  | Truncated cone (American) |
| Commonly used | Europe, Asia (including India) | USA, Canada               |

Projection Symbols



First Angle



Third Angle

Standard projection symbols: First Angle (concentric circles) and Third Angle (truncated cone).

 **Tip:** In India, the First Angle Projection method is standard for engineering drawings. Always check the projection symbol on the drawing sheet.

**Malayalam support:** പ്രോജക്ഷൻ (Projection) — 3D വസ്തുവിനെ 2D പ്രതലത്തിൽ പ്രതിബിംബിപ്പിക്കുന്ന പ്രക്രിയ. പ്രോജക്ഷൻ രീതികൾ (പ്രൊജക്ഷൻ സിസ്റ്റങ്ങൾ), പ്രൊജക്ഷൻ പ്രതലം (പ്രൊജക്ഷൻ പ്ലേൻ) എന്നിവ ഉൾപ്പെടുന്നു. പ്രൊജക്ഷൻ രീതികൾ പ്രകാരം പ്രൊജക്ഷൻ പ്രതലം പ്രതിബിംബിപ്പിക്കുന്നു.

2.2 Projections of Points & Lines▼

**Projection of Points:** A point is a zero-dimensional element. Its projection is a point on the reference plane. The position of the point relative to the Horizontal Plane (HP) and Vertical Plane (VP) determines its projection.

- **First quadrant:** Point is above HP and in front of VP.
- **Second quadrant:** Point is above HP and behind VP.
- **Third quadrant:** Point is below HP and behind VP.
- **Fourth quadrant:** Point is below HP and in front of VP.

**Projection of Lines:** A line is defined by two points. Its projections are obtained by projecting the two endpoints.

- **Parallel to both planes:** True length appears in both views.
- **Parallel to one plane, inclined to the other:** One view shows true length, the other shows foreshortened length.
- **Inclined to both planes:** Both views show foreshortened lengths.

Projection of Lines — Key Cases (First Quadrant)

| Line Position | Front View | Top View |
|---------------|------------|----------|
|---------------|------------|----------|

| Line Position                  | Front View               | Top View                 |
|--------------------------------|--------------------------|--------------------------|
| Parallel to HP & VP            | True length (horizontal) | True length (horizontal) |
| Parallel to HP, inclined to VP | Foreshortened            | True length              |
| Parallel to VP, inclined to HP | True length              | Foreshortened            |
| Inclined to both HP & VP       | Foreshortened            | Foreshortened            |

WORKED EXAMPLE — PROJECTION OF A LINE

**Given:** Line AB, length 60 mm, parallel to HP and inclined 30° to VP. End A is 20 mm above HP and 15 mm in front of VP.

**Front View:** Project A' at 20 mm above XY. Draw a line of length  $60 \times \cos 30^\circ = 51.96$  mm at the given angle.

**Top View:** Project A at 15 mm below XY. Draw the true length 60 mm at 30° to the reference line.

 **Tip:** Always start with the projection of the endpoint that has the most constraints. Use construction lines lightly in pencil. The top view shows the line's inclination to HP; the front view shows inclination to VP.

**Malayalam support:** ലൈൻ AB ന്റെ ദൈർഘ്യം 60 മില്ലീമീറ്റർ ആണ്. ഇത് ഹോറിസന്റൽ പ്ലേൻ (HP) ന്റെ സമാന്തരമായി വെർട്ടിക്കൽ പ്ലേൻ (VP) ന്റെ 30° കോണിൽ ചേർന്നിരിക്കുന്നു. A എന്ന അറ്റം HP-യിൽ 20 മില്ലീമീറ്റർ മുകളിലും VP-യിൽ 15 മില്ലീമീറ്റർ മുന്നിലും സ്ഥിതി ചെയ്യുന്നു. **ഫ്രണ്ട് വ്യൂ:** XY രേഖയിൽ 20 മില്ലീമീറ്റർ മുകളിൽ A' എന്ന ബിന്ദു കണ്ടെത്തുക. 60 × cos 30° = 51.96 മില്ലീമീറ്റർ നീളത്തിൽ 30° കോണിൽ വരയ്ക്കുക. **ടോപ്പ് വ്യൂ:** XY രേഖയിൽ 15 മില്ലീമീറ്റർ താഴെ A എന്ന ബിന്ദു കണ്ടെത്തുക. 60 മില്ലീമീറ്റർ നീളത്തിൽ 30° കോണിൽ വരയ്ക്കുക.

2.3 Orthographic Projections & Sectional Views of Objects▼

**Orthographic Projection:** The representation of a 3D object using multiple 2D views — typically front view, top view, and side view. Each view shows the object as seen from a specific direction.

- **Front View:** Projected on the Vertical Plane (VP).
- **Top View:** Projected on the Horizontal Plane (HP).
- **Side View:** Projected on the Profile Plane (PP).

**Sectional Views:** When an object has internal features (holes, cavities, ribs), a **sectional view** is used to show the internal details. The object is imagined as cut by a cutting plane, and the portion in front of the cutting plane is removed.

- **Cutting plane:** A phantom line (chain line with thick ends) showing where the cut is made.
- **Section lines (hatching):** Thin equidistant lines drawn at 45° on the cut surface.
- **Types:** Full section, half section, offset section, revolved section.

 **Tip:** In orthographic projection, align the views properly — front view above top view, and side view to the side. Use hidden lines (dashed) for edges that are not visible in that view.

 **Common Mistake:** Forgetting to draw hidden lines in orthographic views, or drawing section lines (hatching) on non-sectioned areas. Also, cutting plane lines should be clearly shown with arrowheads indicating the direction of viewing.

**Malayalam support:** മെറ്റീരിയലുകളുടെ ഓരോ ഭാഗത്തും 3D മോഡലിംഗ്, ഓർത്തോഗ്രാഫിക് വ്യൂ, ടോപ്പ് വ്യൂ, സൈഡ് വ്യൂ 2D മോഡലിംഗ് ഉപയോഗിക്കുന്നു. മെറ്റീരിയൽ വ്യൂ (Sectional view) മെറ്റീരിയലുകളുടെ ഓരോ ഭാഗത്തും മെറ്റീരിയലുകളുടെ ഓരോ ഭാഗത്തും.

 **Module II Summary:** First-angle projection is used in India. Points are projected based on their quadrant. Lines show true length when parallel to a plane. Orthographic views (front, top, side) describe 3D objects. Sectional views reveal internal features using a cutting plane and hatching.

MODULE III

# Lab — CAD Software Familiarization & Basic Commands

Learn to use CAD software — open/save drawings, draft settings, draw commands, modify commands, dimensioning, and text.

Hours: 9

CO: CO3

Bloom: Apply

Type: CAD Lab

## 3.1 Introduction to CAD — Interface, Commands & Settings▼

**Computer-Aided Drafting (CAD):** The use of computer software to create, modify, analyse, and optimise designs. CAD replaces manual drafting with digital tools that improve accuracy, efficiency, and ease of modification.

### Advantages of CAD over Manual Drafting

- **Accuracy:** CAD allows precise measurements and alignment.
- **Efficiency:** Repeatable commands, copy-paste, and libraries save time.
- **Modification:** Edit drawings easily without redrawing.
- **Storage:** Digital files are easy to store, share, and version-control.
- **Standardisation:** Layers, linetypes, and text styles ensure consistency.

### Basic CAD Commands

#### Common CAD Commands

| Category | Command | Purpose |
|----------|---------|---------|
|----------|---------|---------|

| Category | Command     | Purpose                                                   |
|----------|-------------|-----------------------------------------------------------|
| Draw     | LINE        | Draw straight line segments                               |
|          | CIRCLE      | Draw a circle (centre + radius / diameter)                |
|          | RECTANGLE   | Draw a rectangle by specifying corners                    |
|          | POLYGON     | Draw a regular polygon (specify sides and centre)         |
| Modify   | ERASE       | Delete selected objects                                   |
|          | COPY        | Duplicate objects                                         |
|          | MIRROR      | Create a mirrored copy                                    |
|          | OFFSET      | Create parallel copies at a specified distance            |
|          | TRIM        | Cut objects at intersections                              |
| Settings | GRID / SNAP | Snap to grid points for precise placement                 |
|          | OSNAP       | Object snap — snap to endpoints, midpoints, centres, etc. |
|          | DIMENSION   | Add dimensions (linear, aligned, radius, diameter)        |

CAD Workflow

- 1. **Setup:** Set units, limits, grid, and snap settings.
- 2. **Draw:** Use draw commands to create the geometry.
- 3. **Modify:** Edit using modify commands.
- 4. **Annotate:** Add dimensions, text, and notes.
- 5. **Save:** Save the drawing as a .dwg file or export as PDF.

 **Tip:** Use keyboard shortcuts for common commands (e.g., L for LINE, C for CIRCLE, E for ERASE, TR for TRIM). This speeds up your workflow significantly.

**Malayalam support:** CAD (Computer-Aided Drafting) — കമ്പ്യൂട്ടർ സഹിതം ഡ്രാഫ്റ്റിംഗ്. LINE, CIRCLE, RECTANGLE വരയ്ക്കാനും, ERASE, COPY, MIRROR, TRIM മാറ്റിക്കൊണ്ടും ഉപയോഗിക്കുന്നു.

 **Module III Summary:** CAD software replaces manual drafting with digital tools. Key commands include LINE, CIRCLE, RECTANGLE, POLYGON (draw), and ERASE, COPY, MIRROR, OFFSET, TRIM (modify). Settings like GRID, SNAP, OSNAP help with precision. Dimensioning and text add annotations.

MODULE IV

Lab — Orthographic & Sectional Views in CAD

Develop orthographic projections and sectional views of engineering objects using CAD software.

4.1 Creating Orthographic & Sectional Views in CAD

This module applies the manual drawing skills from Module II using CAD software. The same principles of orthographic and sectional views are used, but with digital tools.

Step-by-Step Procedure in CAD

- 1. **Analyse the object:** Identify the principal views (front, top, side) and the cutting plane for sectional views.
- 2. **Set up layers:** Create layers for construction lines, object lines, hidden lines, centre lines, section lines, and dimensions.
- 3. **Draw the front view:** Use LINE, CIRCLE, RECTANGLE, and other draw commands to create the front view geometry.
- 4. **Project to top view:** Use projection lines (using OFFSET, COPY, or construction lines) to create the top view aligned below the front view.
- 5. **Project to side view:** Similarly, project the side view from the front view.
- 6. **Add hidden lines:** Use dashed linetypes for edges not visible in each view.
- 7. **For sectional views:** Draw the cutting plane line, then hatch the cut area using the HATCH command with 45° lines.
- 8. **Dimension and annotate:** Add dimensions and text notes.
- 9. **Plot/Print:** Set up the layout, scale, and plot to PDF or print.

CAD Layers for Engineering Drawings

| Layer Name   | Colour      | Linetype                      | Lineweight | Purpose                       |
|--------------|-------------|-------------------------------|------------|-------------------------------|
| Construction | Gray        | Continuous                    | 0.25 mm    | Construction/projection lines |
| Object       | Black/White | Continuous                    | 0.50 mm    | Visible edges                 |
| Hidden       | Red         | Dashed                        | 0.25 mm    | Hidden edges                  |
| Centre       | Blue        | Centre (long dash-short dash) | 0.25 mm    | Centre lines                  |
| Section      | Green       | Continuous                    | 0.25 mm    | Section/hatch lines           |
| Dimension    | Cyan        | Continuous                    | 0.25 mm    | Dimensions and text           |

**Important:** Always use layers for different line types. This allows you to control visibility, plot styles, and edit efficiently. Do not draw everything on a single layer.

**Tip:** Use the HATCH command with the "ANSI31" pattern (45° lines) for section hatching. Adjust the scale to get the right spacing. For CAD orthographic views, use PROJECT or construction lines to maintain alignment between views.

**Malayalam support:** CAD-ഓഫ് ലെയറുകൾ ഉപയോഗിക്കുക, ലെയറുകൾ വ്യത്യസ്തമായി — വരകൾ വരയ്ക്കുകയും പരസ്പരം തുല്യമായി വരയ്ക്കുകയും ചെയ്യുക. ലെയറുകൾ വ്യത്യസ്തമായി വരയ്ക്കുകയും പരസ്പരം തുല്യമായി വരയ്ക്കുകയും ചെയ്യുക. HATCH കമാന്റ് ഉപയോഗിച്ച് 45° ലൈനുകളിൽ ഷാഡ് ചെയ്യുക.

 **Module IV Summary:** Use CAD software to create orthographic views (front, top, side) and sectional views. Organise drawing elements using layers (Object, Hidden, Centre, Section, Dimension). Use HATCH for section lines. Project views using construction lines to maintain alignment.

## Formula Bank

### Regular Polygon — Interior Angle

$$(n - 2) \times 180^\circ / n$$

n = number of sides

### Regular Polygon — Exterior Angle

$$360^\circ / n$$

n = number of sides

### Circumradius (Polygon)

$$R = s / (2 \times \sin(180^\circ/n))$$

s = side length, n = number of sides

### Apothem (Polygon)

$$r = s / (2 \times \tan(180^\circ/n))$$

s = side length, n = number of sides

### Hexagon — Circumradius

$$R = s$$

Side length equals circumradius

### Cylinder — Lateral Surface Area

$$2\pi rh$$

r = radius, h = height

### Cylinder — Total Surface Area

$$2\pi r(r + h)$$

r = radius, h = height

### Cylinder — Circumference

$$2\pi r$$

Used for cylinder development

## Diagram Library for Rapid Revision



### **Regular Polygons**

Module I — Pentagon and Hexagon construction using circumcircle method.

### **Conic Sections**

Module I — Ellipse using concentric circle method.

### **Surface Development**

Module I — Nets for rectangular prism and cylinder.

### **Projection Symbols**

Module II — First-angle and third-angle projection symbols.

### **Orthographic Views**

Module II — Front, top, and side view alignment.

### **Sectional Views**

Module II — Cutting plane and hatching for sectional views.

## Self-Learning Activities (Official)

### **Module I — Regular Polygon Terminology**

1.5 hours

Definition and terminology of regular polygons. Construction of pentagon, hexagon, heptagon, octagon with given side length.

### **Module I — Conic Sections**

1.5 hours

Methodology and terminology of conic sections. Construction of ellipse, parabola, and hyperbola.

### **Module I — Development of Surfaces**

1.5 hours

Procedure familiarisation for engineering applications of surface development.

### **Module II — Theory of Projection**

1.5 hours

Outline the theory of projection, first-angle and third-angle projections. Projection of points and lines.

### **Module II — Orthographic & Isometric**

3 hours

Practice more exercises on orthographic and isometric projection.

### **Module II — Sectional Views**

3 hours

Draw sectional views of various simple engineering objects. Practice more exercises.

### **Module III — CAD Basics**

1.5 hours

Introduction to Computer Aided Drafting: applications, advantages, hardware/software requirements, CAD basics.

### **Module III — Draw & Modify Commands**

1.5 hours

Limits of drawing, units, grid, snap, osnap, co-ordinates, ortho mode. Draw and modify commands.

### **Module III — Properties & Dimensioning**

1.5 hours

Working with CAD: properties of lines (colour, line weight, line type), dimensions and text on drawings.

### **Module IV — Orthographic Views in CAD**

1.5 hours

Developing orthographic views of simple components with dimensions and text. More practice exercises.

### **Module IV — Sectional Views in CAD**

1.5 hours

Draw sectional views of simple engineering objects. More practice exercises.

### **Series Exam Preparation**

1.5 hours

Series exam preparation (Modules I & II / Modules III & IV).

**Total Self-Learning Hours:** 22.5

## Assessment Methodology

### Assessment Breakdown — Lab

| Mode | Attendance | CE | CA1 | CA2 | OEE | ESE | Total |
|------|------------|----|-----|-----|-----|-----|-------|
|------|------------|----|-----|-----|-----|-----|-------|

| Mode                                                                                         | Attendance | CE | CA1 | CA2 | OEE | ESE | Total |
|----------------------------------------------------------------------------------------------|------------|----|-----|-----|-----|-----|-------|
| Continuous evaluation                                                                        | —          | 10 | 15  | 15  | 10  | —   | —     |
| Practical test (first half)                                                                  |            |    |     |     |     |     |       |
| Practical test (second half)                                                                 | —          | —  | —   | —   | —   | —   | —     |
| Open-ended experiment                                                                        | —          | —  | —   | —   | —   | —   | —     |
| Practical (Duration 3h)                                                                      | —          | —  | —   | —   | —   | 40  | 40    |
| Converted to                                                                                 | —          | 5  | 5   | 10  | —   | —   | —     |
| Total marks                                                                                  | —          | 15 | 15  | 10  | —   | 40  | 100   |
| CIE: 60 marks (Attendance 5 + CE 15 + CA1 15 + CA2 15 + OEE 10)   ESE: 40 marks   Total: 100 |            |    |     |     |     |     |       |

### Scheme of Evaluation — Practical (CIE)

- **Preparation & Punctuality:** 10 marks
- **Experimental Setup & Procedure:** 10 marks
- **Observation & Data Recording:** 5 marks
- **Analysis & Result Interpretation:** 10 marks
- **Viva Voce / Concept Understanding:** 10 marks
- **Workmanship, Discipline & Teamwork:** 5 marks

**Total: 50 marks**

### Scheme of Evaluation — ESE (Practical)

- **Write-up / Procedure:** 10 marks
- **Experiment Setup & Execution:** 10 marks
- **Observation & Result / Output:** 8 marks
- **Viva Voce:** 8 marks
- **Record:** 4 marks

**Total: 40 marks**

### Open-Ended Experiment (OEE)

- **Originality & Relevance:** 10 marks
- **Clarity of Objectives & Plan:** 10 marks
- **Correctness of Execution & Recording:** 10 marks
- **Analysis, Interpretation & Presentation:** 10 marks
- **Teamwork & Innovation:** 10 marks

**Total: 50 marks**

## Series Test & ESE Pattern

- **Series Test I:** Modules I & II (1.5 hrs)
- **Series Test II:** Modules III & IV (1.5 hrs in CAD)
- **ESE Split-up:**
  - 10 MCQs/Fill-ups (CO1 & CO2) — 10 marks
  - CAD Drawing (CO3 & CO4) — 18 marks
  - Viva Voce — 8 marks
  - Record — 4 marks

**ESE Total: 40 marks**

 **Note:** Students need to draw exercises from Module II on A2/A4 size sheets. If using A4, include sheet size, margin, title block, and angle of projection. The Module II sheets may be considered as the open-ended project. End semester examination is conducted in CAD only. Students must submit a certified fair record for ESE with figures and construction procedure (Module II, Part A) and CAD drawings (Modules III & IV, Part B).

## Module-wise Questions with Answers

### Module I — Regular Polygons, Conic Sections, Surface Development

**Q1: What is a regular polygon? How do you construct a regular hexagon given the side length?▼**

**Answer:** A regular polygon is a closed plane figure with all sides equal and all interior angles equal.

**Construction of hexagon (side length s):**

1. Since  $R = s$  for a hexagon, draw a circle of radius  $s$ .
2. Mark any point on the circle as the first vertex.
3. Using the same radius  $s$ , step off 6 equal divisions around the circle ( $60^\circ$  each).
4. Join the consecutive points to form the hexagon.

**Formula:** Interior angle =  $(6-2) \times 180^\circ / 6 = 120^\circ$ . Exterior angle =  $360^\circ / 6 = 60^\circ$ .

**Q2: Explain the concentric circle method for constructing an ellipse.▼**

**Answer:** The concentric circle method is used to construct an ellipse when the major and minor axes are given.

1. Draw two concentric circles with diameters equal to the major axis (outer circle) and minor axis (inner circle).
2. Divide the circles into 12 equal parts ( $30^\circ$  intervals) using radial lines from the centre.
3. From each intersection on the outer circle, draw a vertical line downward.

4. From each intersection on the inner circle, draw a horizontal line outward.
5. The intersections of these vertical and horizontal lines give points on the ellipse.
6. Join the points with a smooth curve.

**Key principle:** The ellipse is the locus of points whose sum of distances to two foci is constant.

### Q3: What is meant by "development of surfaces"? Draw the development of a cylinder.▼

**Answer:** Development of a surface is the process of unfolding a 3D solid's surface into a 2D plane (a net) without distortion.

#### Cylinder development:

- **Lateral surface:** A rectangle with height = cylinder height ( $h$ ) and width = circumference ( $2\pi r$ ).
- **Bases:** Two circles of radius  $r$  attached to the top and bottom edges of the rectangle.

**Area formulas:** Lateral =  $2\pi rh$ , Total =  $2\pi r(r + h)$ .

**Application:** Used in sheet metal work to determine the material needed to make a cylindrical object.

## Module II — Projections, Orthographic & Sectional Views

### Q4: Distinguish between first-angle and third-angle projection.▼

#### Answer:

- **First-angle projection:** Object is placed between the observer and the projection plane. The front view is above the top view. Used in Europe and India. Symbol: concentric circles.
- **Third-angle projection:** Object is placed between the projection plane and the observer. The front view is below the top view. Used in USA and Canada. Symbol: truncated cone.

In India, the **first-angle projection** method is standard.

### Q5: How do you project a line that is parallel to HP and inclined to VP?▼

**Answer:** For a line parallel to HP and inclined to VP:

- **Top view (on HP):** Shows the true length of the line, since the line is parallel to HP.
- **Front view (on VP):** Shows the foreshortened length ( $\text{length} \times \cos \theta$ ), where  $\theta$  is the angle of inclination to VP.

**Procedure:** First draw the top view with the true length at the given angle to the reference line (XY). Then project the front view vertically, showing the foreshortened length.

#### Q6: What is a sectional view? Why is it used in engineering drawing?▼

**Answer:** A sectional view is a drawing that shows the internal features of an object by imagining it cut by a cutting plane. The portion in front of the cutting plane is removed to reveal the interior.

**Uses:**

- Reveals hidden internal details (holes, cavities, ribs).
- Clarifies complex assemblies.
- Reduces the need for hidden lines.

**Key elements:** Cutting plane line (phantom line with arrowheads), section lines (hatching at 45°) on the cut surface.

## Module III & IV — CAD Software

#### Q7: List the advantages of CAD over manual drafting.▼

**Answer:**

- **Accuracy:** Precise measurements and alignment.
- **Efficiency:** Commands like COPY, ARRAY, MIRROR save time.
- **Modification:** Easy to edit without redrawing.
- **Storage:** Digital files are compact and easy to share.
- **Standardisation:** Layers and templates ensure consistency.
- **Reusability:** Components can be reused from libraries.
- **Visualisation:** 3D modelling and rendering capabilities.

#### Q8: Explain the use of layers in CAD. Name the common layers used in engineering drawing.▼

**Answer:** Layers in CAD are like transparent overlays that organise different types of drawing elements. Each layer can have its own colour, linetype, and linewidth.

**Common layers:**

- **Construction:** Construction/projection lines (gray, continuous, thin).
- **Object:** Visible edges (black/white, continuous, thick).
- **Hidden:** Hidden edges (red, dashed).
- **Centre:** Centre lines (blue, long dash-short dash).
- **Section:** Section/hatch lines (green, continuous).

- **Dimension:** Dimensions and text (cyan, continuous).

Using layers makes editing, plotting, and visibility control easier.

#### Q9: Describe the procedure to create a sectional view in CAD.▼

##### Answer:

1. Draw the orthographic view of the object.
2. Draw the cutting plane line (phantom line) with arrowheads showing the direction of viewing.
3. Use the TRIM or EXTEND commands to prepare the cut geometry.
4. Use the HATCH command to apply hatching (45° lines, ANSI31 pattern) to the cut surface.
5. Add dimensions and text to complete the sectional view.
6. Place the sectional view on the drawing layout with proper alignment.

**Tip:** Use a separate layer for the hatch pattern so it can be turned off if needed.

#### Q10: What are the key settings to configure before starting a CAD drawing?▼

##### Answer:

- **Units:** Set to millimeters (mm) for engineering drawings.
- **Limits:** Define the drawing boundary.
- **Grid & Snap:** Enable grid display and snap to grid points for precision.
- **Object Snap (OSNAP):** Enable endpoints, midpoints, centres, intersections, etc.
- **Layers:** Create and assign layers for different line types.
- **Text Style:** Set a standard text height and font.
- **Dimension Style:** Set arrowheads, text offset, and precision.

Proper initial setup saves time and ensures consistency throughout the drawing.

## Practice Model Paper

*Based on the official pattern. This is a practice paper, not an official SITTTTR model question paper.*

### Course 1004 — Engineering Drawing with CAD

#### End Semester Examination (ESE) — Practice Paper

**Duration:** 3 hours | **Total Marks:** 40

#### Part A — Theory (CO1 & CO2)

Answer any **5** of the following. Each question carries 2 marks. ( $5 \times 2 = 10$  marks)

1. Define a regular polygon. Write the formula for its interior angle.
2. Explain the concentric circle method for constructing an ellipse.
3. What is meant by the development of surfaces? Give one engineering application.
4. Distinguish between first-angle and third-angle projection with sketches.
5. How do you project a line that is parallel to VP and inclined to HP?
6. Why are sectional views used in engineering drawing?

### Part B — CAD Drawing (CO3 & CO4)

Answer any **1** of the following. Each question carries 18 marks. ( $1 \times 18 = 18$  marks)

1. Using CAD software, draw the orthographic views (front, top, and side) of a given isometric object (Figure provided in the exam). Show all dimensions and hidden lines.
2. Using CAD software, draw the sectional view of a given engineering component, showing the cutting plane and hatching. Include all dimensions.

### Part C — Viva Voce & Record

- **Viva Voce:** 8 marks — Questions based on concepts from all modules.
- **Record:** 4 marks — Completion and neatness of practical record, certified by faculty.

**Total:**  $10 + 18 + 8 + 4 = 40$  marks

## Quick Revision

### Module I — Polygons, Conics, Development

- Regular polygon: all sides & angles equal.
- Interior angle =  $(n-2) \times 180^\circ / n$ ; Exterior =  $360^\circ / n$ .
- Hexagon:  $R = s$  (circumradius = side length).
- Ellipse: concentric circle method uses 2 circles (major/minor axes).
- Development: unfolding 3D  $\rightarrow$  2D net.
- Cylinder development: rectangle ( $2\pi r \times h$ ) + 2 circles.

### Module II — Projections & Views

- First-angle: object between observer & plane (India).
- Third-angle: plane between object & observer (USA).
- Line parallel to HP  $\rightarrow$  true length in top view.
- Line parallel to VP  $\rightarrow$  true length in front view.
- Orthographic views: front, top, side.
- Sectional views: cutting plane + hatching ( $45^\circ$ ).

### Module III — CAD Basics

- CAD: Computer-Aided Drafting — digital drawing.
- Advantages: accuracy, efficiency, easy modification.



- Draw commands: LINE, CIRCLE, RECTANGLE, POLYGON.
- Modify commands: ERASE, COPY, MIRROR, OFFSET, TRIM.
- Settings: GRID, SNAP, OSNAP, UNITS, LIMITS.
- Dimensioning: linear, aligned, radius, diameter.

### Module IV — CAD Views

- Use layers: Object, Hidden, Centre, Section, Dimension.
- Project views using construction lines.
- HATCH command for section lines (ANSI31 pattern).
- Apply dimensions and text annotations.
- Plot/print with proper scale and layout.
- Save as .dwg or export as PDF.

### Common Mistakes to Avoid

- ❌ Forgetting hidden lines in orthographic views.
- ❌ Using the wrong projection method symbol on the drawing sheet.
- ❌ Incorrect hatching (section lines drawn on non-cut areas).
- ❌ Drawing all CAD elements on a single layer (no organisation).
- ❌ Forgetting to set units and limits before starting a CAD drawing.
- ❌ Misaligning front, top, and side views in orthographic projection.
- ❌ Using the wrong formula for polygon circumradius (especially hexagon).

### Exam Checklist

- ✅ Know the projection symbols (First-angle vs Third-angle).
- ✅ Practice constructing pentagon and hexagon with given side length.
- ✅ Be able to draw ellipse using both rectangular and concentric circle methods.
- ✅ Understand surface development for prism and cylinder.
- ✅ Practice orthographic projection of simple objects (front, top, side).
- ✅ Practice sectional views with cutting plane and hatching.
- ✅ Know CAD commands (draw, modify, settings, dimensioning).
- ✅ Use layers properly in CAD drawings.
- ✅ Prepare a neat record with all exercises and construction procedures.

## Glossary

### Key Terms and Meanings

| Term            | Meaning                                                                            |
|-----------------|------------------------------------------------------------------------------------|
| Regular Polygon | A closed plane figure with all sides and angles equal.                             |
| Circumradius    | Radius of the circle passing through all vertices of a polygon.                    |
| Apothem         | Distance from the centre to the midpoint of a side of a polygon.                   |
| Conic Section   | Curve obtained by intersecting a cone with a plane (ellipse, parabola, hyperbola). |

| Term                          | Meaning                                                                                   |
|-------------------------------|-------------------------------------------------------------------------------------------|
| Ellipse                       | Conic section where the plane cuts all generators of the cone.                            |
| Development (of a surface)    | Unfolding a 3D surface into a 2D plane (net).                                             |
| Orthographic Projection       | Representing a 3D object using multiple 2D views from perpendicular directions.           |
| First-Angle Projection        | Projection method where the object is between the observer and the plane (used in India). |
| Third-Angle Projection        | Projection method where the plane is between the observer and the object (used in USA).   |
| Sectional View                | A view showing internal details by cutting the object with a plane.                       |
| Cutting Plane                 | A phantom line showing where the object is cut for a sectional view.                      |
| Hatching                      | Thin 45° lines drawn on the cut surface in a sectional view.                              |
| CAD (Computer-Aided Drafting) | Software used to create and modify digital engineering drawings.                          |
| Layer                         | An organisational tool in CAD that groups drawing elements with common properties.        |
| OSNAP (Object Snap)           | CAD setting that snaps to specific points (endpoint, midpoint, centre, etc.).             |
| HATCH                         | CAD command used to fill an area with a pattern (e.g., section lines).                    |

## Learning Resources

### Textbooks

#### Recommended Textbooks

| Sl. No. | Title                                              | Author                      | Publication                  |
|---------|----------------------------------------------------|-----------------------------|------------------------------|
| 1       | Engineering Graphics with AUTOCAD                  | P I Varghese                | VIP Publishers               |
| 2       | Engineering Graphics                               | K. C John                   | PHI Learning Private Limited |
| 3       | Engineering Graphics unique methods easy solutions | K N Anilkumar               | Adhyuth Narayan Publishers   |
| 4       | Engineering Drawing with CAD Applications          | N. D. Bhatt & V. M. Panchal | Charotar Publishing House    |

### References

#### Reference Books

| Sl. No. | Title                             | Author                                   | Publication                               |
|---------|-----------------------------------|------------------------------------------|-------------------------------------------|
| 1       | Engineering Drawing               | N. D. Bhatt                              | Charotar Publishing House, Anand, Gujarat |
| 2       | Engineering Graphics with AutoCAD | D.M. Kulkarni, A.P. Rastogi, A.K. Sarkar | PHI Learning Private Limited              |
| 3       | Engineering Drawing               | M. B. Shah and B.C. Rana                 | Pearson Publications                      |

### Web Resources

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**Major Equipment (for a batch of 30 students):** 30 Desktop Computers (Intel i3/i5, 8GB RAM, 500GB HDD/SSD, Ubuntu 22.04 / Windows 10).

**Major Software:** Suitable version of CAD software (open access/licensed) — 30 licenses.